

Operating System Security

1. Protection

Protection refers to a mechanism that controls the access of programs, processes, or users to the resources defined by a computer system. We can take protection as a helper to multiprogramming operating systems so that many users might safely share a common logical namespace such as a directory or files.

2. What is Operating System Security?

Measures to prevent a person from illegally using resources in a computer system, or interfering with them in any manner. These measures ensure that data and programs are used only by authorized users and only in a desired manner, and that they are neither modified nor denied to authorized users.

Security measures deal with threats to resources that come from outside a computer system, while protection measures deal with internal threats. Passwords are the principal security tool.

3. Goal — CIA Triad

The CIA triad represents the three core goals of information security:

Principle	Meaning
Confidentiality	Data should be accessible only to authorized users.
Integrity	Data and programs should be protected from unauthorized modification.
Availability	Authorized users should not be denied access to required resources.

4. Security Can Be Attacked in the Following Ways

No.	Attack / Weakness
1	Authorization
2	Browsing
3	Trap doors
4	Invalid Parameters
5	Line Tapping
6	Electronic Data Capture
7	Lost Line
8	Improper Access Controls
9	Waste Recovery
10	Rogue Software

5. Threats to Operating System

5.1 Malware

Malware is short for malicious software and refers to any software that is designed to cause harm to computer systems, networks, or users. Malware can take many forms. Malware is a program designed to gain access to computer systems, generally for the benefit of some third party, without the user's permission.

5.2 Network Intrusion

A system called an Intrusion Detection System (IDS) observes network traffic for malicious transactions and sends immediate alerts when it is observed. It is software that checks a network or system for malicious activities or policy violations.

Each illegal activity or violation is often recorded either centrally using a SIEM system or notified to an administration.

5.3 Buffer Overflow Technique

The buffer overflow technique can be employed to force a server program to execute an intruder-supplied code to breach the host computer system's security. It has been used to a devastating effect in mail servers and other Web servers.

The basic idea in this technique is simple. Most systems contain a fundamental vulnerability—some programs do not validate the lengths of inputs they receive from users or other programs.

Because of this vulnerability, a buffer area in which such input is received may overflow and overwrite contents of adjoining areas of memory. On hardware platforms that use stacks that grow downward in memory, e.g., the Intel 80x86 architecture, such overflows provide an opportunity to execute a piece of code that is disguised as data put in the buffer. This code could launch a variety of security attacks.

6. How a Buffer Overflow Can Be Used to Launch a Security Attack

- The stack grows downward, i.e., toward smaller addresses in memory. It looks as shown on the left before the currently executing function calls the function *sample*.
- The code of the calling function pushes a return address and two parameters of *sample* onto the stack. Each of these occupies four bytes.
- The code of *sample* allocates the variable *beta* and other variables on the stack. The stack now looks as shown on the right. Notice that the start address of *beta* is at the low end of the memory allocated to it. The end address of *beta* adjoins the last byte of the parameters.
- The function *sample* copies 412 bytes into the variable *beta*. The first 408 bytes contain code whose execution would cause a security violation. Bytes 409–412 contain the start address of this code. These four bytes overwrite the return address in the stack.
- The function *sample* executes a return statement. Control is transferred to the address found in the stack entry that is expected to contain the return address. Effectively, the code in variable *beta* is invoked. It executes with the privileges of the calling function.

7. Types of Threats

7.1 Program Threats

Threat	Description
Virus	A virus is a malicious executable code attached to another executable file. The virus spreads when an infected file is passed from system to system. Viruses can be harmless or they can modify or delete data. Opening a file can trigger a virus.
Trojan Horse	A Trojan horse is malware that carries out malicious operations under the appearance of a desired operation such as playing an online game.
Logic Bomb	A logic bomb is a malicious program that uses a trigger to activate the malicious code. The logic bomb remains non-functioning until that trigger event happens.

7.2 System Threats

Threat	Description
Worm	Worms replicate themselves on the system, attaching themselves to different files and looking for pathways between computers, such as computer network that shares common file storage areas.
Denial of Service	Denial of Service (DoS) is a cyber-attack on an individual Computer or Website with the intent to deny services to intended users. Their purpose is to disrupt an organization's network operations by denying access to its users.

8. How to Ensure Operating System Security?

8.1 Authorization

It means verification of access to the system resources. Intruders may guess or steal password and use it. Intruder may use a vendor-supplied password, which is expected to use by system administrator. It may find password by trial and error method.

If the user logs on and goes for a break then the intruder may use the terminal. An intruder can write a dummy login program to fool user and that program collects information for its use later on.

8.2 Authentication

Authentication is verification of a user's identity. Operating systems most often perform authentication by knowledge. That is, a person claiming to be some user X is called upon to exhibit some knowledge shared only between the OS and user X, such as a password.

8.3 Browsing

Files are very permissive so one can easily browse system files. Due to that it may access database and confidential information can be read.

8.4 Trap Doors

Sometimes Software designers want to modify their programs after installation. for that there are some secret entry points which programmers keep and it does not require any permission. These are called trap doors. Intruders can use these trap doors.

8.5 Invalid Parameters

Due to invalid parameters some security violation can take place.

8.6 Line Tapping

Tapings in the communication line can access or modify confidential data.

8.7 Electronic Data Capture

Using wiretaps or mechanism to pick up screen radiation and recognize what is displayed on screen is termed electronic data capture.

8.8 Lost Line

In networking, the line way gets lost. In such case some o/s log out and allow access only after correct identify of user. some o/s cannot do this. So process will be floating and allow intruder to access data.

8.9 Improper Access Controls

Some administrators may not plan about all rights. So some users may have more access and some users have very less access.

8.10 Waste Recovery

If the block is deleted its information will be as it is. until it is allocated to another file. Intruder may use some mechanism to scan these blocks.

8.11 Rogue Software

Programs are written to create mischief.

9. Quick Revision

Topic	Key Point
Protection	Mechanism that controls access of programs, processes, or users to computer-system resources.
Operating System Security	Measures to prevent illegal use of resources or interference with them.

Topic	Key Point
CIA Triad	Confidentiality, Integrity, and Availability.
Malware	Malicious software designed to cause harm or gain unauthorized access.
Network Intrusion	Malicious activity or policy violation against a network or system.
IDS	Checks a network or system for malicious activities or policy violations and sends alerts.
SIEM	Can centrally record illegal activities or policy violations.
Buffer Overflow	A buffer area may overflow and overwrite adjoining memory when input is not properly validated.
Virus	Malicious executable code attached to another executable file.
Trojan Horse	Malware that performs malicious operations while appearing to perform a desired operation.
Logic Bomb	Malicious program that activates its malicious code when a trigger event occurs.
Worm	Threat that replicates and looks for pathways between computers.
Denial of Service	Attack intended to deny services to intended users.
Authentication	Verification of a user's identity.
Authorization	Verification of access to system resources.
Trap Door	Secret software entry point that does not require normal permission.
Waste Recovery	Deleted information may remain until the block is allocated to another file.